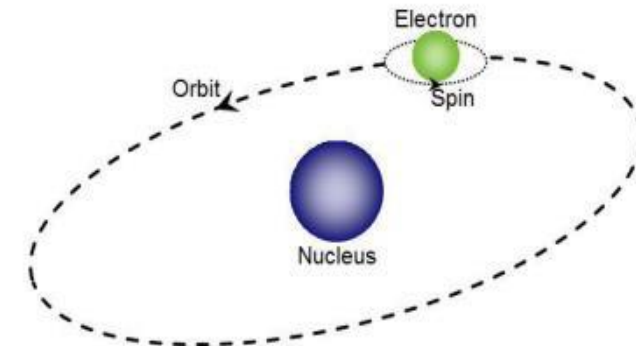


The origin of the magnetic field “magnetism”

Magnetic field is generated due to motion of electrons (electric charges) (charged particles) within its atoms.

- 1) The motion of the electrons in an orbit around the nucleus
- 2) The spin of the electrons around its axis



The magnetic field

Properties:

Imaginary – never cross each other – emerge from the north pole to south pole – into from S to N

Oersted experiment

“Every electric current produces a magnetic field”

We can examine the **shape** of magnetic field by using **iron fillings**

We can determine the **direction** of magnetic field by using **compass**

The magnetic flux

$$\phi_m = BA \cos \theta$$

when the coil is parallel to the magnetic field $\phi_m = \text{zero}$

when the coil is perpendicular to the magnetic field $\phi_m = BA$ “maximum value”

when the coil is making angle with the magnetic field $\phi_m = BA \cos (90 - \theta)$

The magnetic field due to current passing through a conductor

	Stright wire	Circular coil	solenoid
shape	➤ The magnetic lines of around a straight wire carrying current are concentric circles whose centers lie on the wire ➤ The circular magnetic flux lines are closer together near the wire and farther apart from each other as the distance from the wire increases ➤ The current passing through the wire increases, the concentric circles more crowded	➤ The flux lines are circular surrounding the two sides ➤ The flux lines are no longer circular near the center ➤ The flux lines become straight and parallel lines perpendicular to the plane of the coil at the center of the coil	➤ The lines of magnetic flux outside the solenoid are closed coils moving from the North Pole to the South Pole ➤ The lines of magnetic flux through the middle of the solenoid (In a solenoid) are straight and parallel to the axis
direction	➤ Ampere Right Hand Rule.	➤ The right Cork Screw rule ➤ Clockwise rule	➤ Ampere Right Hand Rule. ➤ Clockwise rule
magnitude	$B = \frac{\mu I}{2\pi d}$ This relation is called "Ampere's circular law"	$B = \frac{\mu NI}{2r}$	$B = \frac{\mu NI}{l}$ $B = \mu nI$

The Rules to determine the directions

	Ampere Right Hand Rule	The right Cork Screw rule (Maxwell's Screw Rule)	Clockwise rule (polarity coil)	Fleming left hand rule
The use	To determine the direction of the magnetic field resulting from an electric current in a ➤ straight wire ➤ solenoid	The right Cork Screw rule used to determine the direction of the magnetic field resulting from an electric current in ➤ a circular coil	Used to determine the polarity of ➤ circular coil ➤ solenoid coil	Used to determine the direction of the force (motion) of ➤ straight wire
Description	To straight wire ➤ the thumb points in the direction of the current ➤ the rest of the fingers around the wire will indicate the direction of the magnetic field "To solenoid" ➤ the thumb points in the magnetic field (North pole) ➤ the rest of the fingers around the current direction	➤ the direction of rotation is points in the direction of the current ➤ The direction of motion will give the direction of the magnetic field	➤ The side at which the current is in a clock wise direction is the South Pole ➤ the side of which the current is an antilock wise direction is the North Pole.	➤ The thumb points in the direction of motion of the wire. ➤ The pointer (first finger) points in the direction of the field ➤ The middle (second finger) points in the direction of the current through the wire

The definitions

electromagnetism	Magnetic field is produced around a conductor when an electric current passing through it
The magnetic field	is the area around a magnet (or a current – carrying a conductor) where magnetic forces can act
The Magnetic flux	is a measurement of the total magnetic field which passes through a certain area
The magnetic flux	It is the total number of magnetic flux lines passing through a surface
The magnetic flux density	It is the total number of magnetic flux lines passing normally through a unit area around the point
The magnetic flux density	It is the magnetic force exerts on a wire of 1m length carrying a current of intensity 1 A and placed perpendicularly to the magnetic field
Tesla	It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field
Tesla	It is the magnetic flux density when the total number of magnetic flux lines passing normally through a unit area around the point is 1 Wb
The magnetic permeability	It is the ability of medium to penetrate the magnetic flux lines
Neutral point	It is the point at which the total magnetic flux density vanishes

The factors affecting

	Magnetic flux density Due to straight wire	Magnetic flux density Due to circular coil	Magnetic flux density Due to solenoid	Magnetic force acting on a wire
<i>The factors affecting</i>	➤ intensity of current “directly” ➤ magnetic Permeability of medium “directly” ➤ The distance of the point from the wire “inversely”	➤ intensity of current “directly” ➤ magnetic Permeability of medium “directly” ➤ number of turns on circular coil “directly” ➤ the radius of the circular coil “inversely”	➤ intensity of current “directly” ➤ magnetic Permeability of medium “directly” ➤ number of turns on solenoid “directly” ➤ the length of the solenoid “inversely”	➤ The length of the wire “directly” ➤ The current in the wire “directly” ➤ The magnetic flux density “directly”

The measuring unit

The magnetic flux through a surface	Weber	T.m ²
the magnetic flux density	Tesla	Wb/m ²
The magnetic permeability	Wb/m ²	T.m/A